FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2015

ANSWERS, HINTS & SOLUTIONS
FULL TEST– I
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	B	B
2.	A	A	B
3.	B	B	A
4.	A	D	B
5.	A, B	A, B	A, B, C
6.	B, D	B, D	A, C, D
7.	A, D	A, D	A, B, C, D
8.	A, D	A, C, D	A, B, C
9.	B	D	D
10.	A	B	B
11.	B	B	C
1.	(A) → (p, r, s) (B) → (p, q, r, s, t) (C) → (p, s, t) (D) → (p, s)	(A → p, q, r, s) (B → p, q) (C → p, q, r, s, t) (D → q)	(A) → (r, s) (B) → (p, q) (C) → (r) (D) → (s)
2.	(A) → (p, q, r, s, t) (B) → (p, q, r, s, t) (C) → (p, q, r, s, t) (D) → (p, q, r, s, t)	(A → p, q, t) (B → p, q, s, t) (C → q, r, t) (D → p, q, t)	(A) → (q, s, t) (B) → (p, r) (C) → (s, t) (D) → (q, s)
1.	6	4	3
2.	5	2	4
3.	8	5	6
4.	0	2	6
5.	5	3	1
6.	3	4	6

Physics

PART – I

SECTION – A

7. γ-radiation being neutral is not deflected by magnetic field α & β particle may follow helical path or straight line depending upon velocity of particle & magnetic field.

8. Comparing the eqn. with velocity of projectile

$$\vec{v} = u\cos\theta i + \sqrt{u^2\sin^2\theta - 2gy}\,\hat{j}$$

Acceleration of particle $a_y = \frac{\alpha k_2^2}{2}$, $a_x = 0$

$\therefore$ Motion will be projectile motion and hence the result.

$\therefore$ Force acting on particle will be constant & equal to $\frac{m\alpha k_2^2}{2}$

$\therefore$ $< F > = \frac{m\alpha k_2^2}{2}$

9. $(h_1\rho_1 g)A = (h_2\rho_2 g)A$

$$\frac{h_1}{h_2} = \frac{2}{1}$$

10. $(h_1 + x_1)\rho_1 g = (h_2 + x_2)\rho_2 g$

$x_1/x_2 = 2$ …(i)

$x_1 + x_2 = 3$ …(ii)

$x_1 = 2$ m

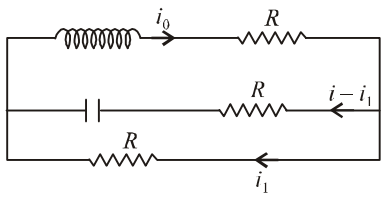
11. $W_1 = \int dW_1 = \int_2^4 (p_0 + x\rho_1 g)Adx = 260$ J

$W_2 = \int dW_2 = \int_1^2 (p_0 + x\rho_2 g)Adx = 130$ J

$W = W_1 + W_2 = 390$ J

SECTION – C

4. $R = \sqrt{\frac{L}{C}} \Rightarrow RC = L/R$ i.e. time constants of RC & LR circuit are same.



After switching from 1 to 2

$$i_0 = \frac{\xi}{2R}$$

from loop rule,

$-(i_0 - i_1)R + \varepsilon/2 + i_1 R = 0$

$\Rightarrow$ $i_1 = \frac{2i_0R - \xi}{4R} = 0$

5. $mgx_2 + \frac{1}{2}kx_2^2 + \frac{1}{2}kx_2^2$

$= \frac{1}{2}k\left(\frac{3mg}{k}\right)^2 - mg\left(\frac{3mg}{k}\right)$

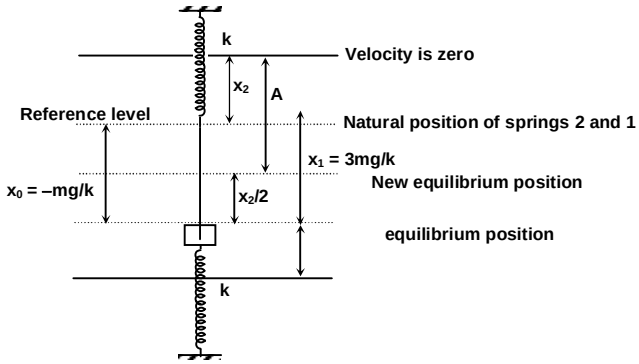
$kx_1^2 + mgx_2 = \frac{9m^2g^2}{2k} - \frac{3m^2g^2}{k} = \frac{3m^2g^2}{2k}$

$2k^2x_2^2 + 2mgkx_2 - 3m^2g^2 = 0$

$x_2 = \frac{-2mgk \pm \sqrt{4m^2g^2k^2 + 24m^2g^2k^2}}{4k^2}$

$= \frac{-2mgk \pm 2mgk\sqrt{7}}{4k^2} = \frac{mg}{2k}[\sqrt{7} - 1]$

$A = \frac{x_0}{2} + x_2 = \frac{mg\sqrt{7}}{2k} = \frac{10 \times \sqrt{7}}{2 \times 100\sqrt{7}} = \frac{1}{20}m = 5cm$



6. $\lambda_0 - V_s t = x_1$

$\lambda_0 + V_s t = x_2$

$V_s = \frac{x_2 - x_1}{2T}$ = 60 m/s & $\lambda_0 = \frac{x_1 + x_2}{2}$ = 1.1 m

Speed of sound V = n × λ_0 = 330 m/s

Frequency detected at point Q

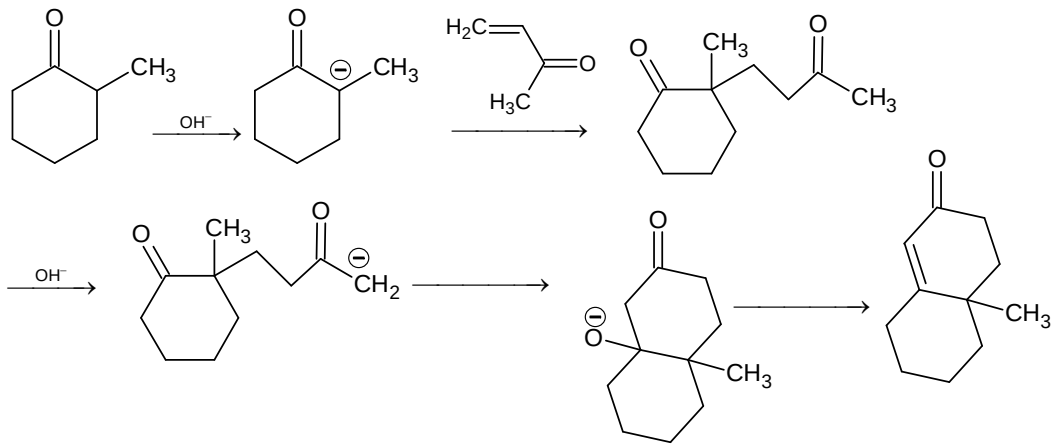
$n = \frac{V}{V - V_s} \times 300\,Hz = 330\,Hz$

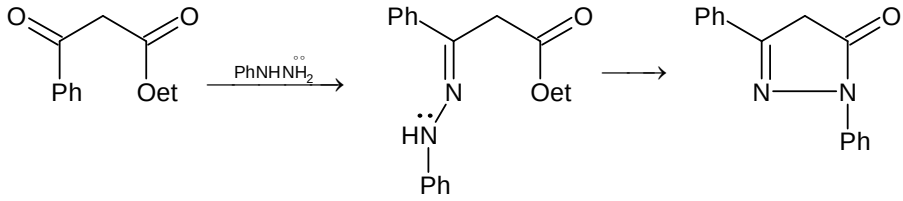
Chemistry

PART – II

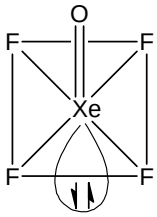
SECTION – A

2.

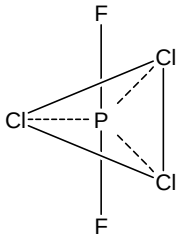


3. 

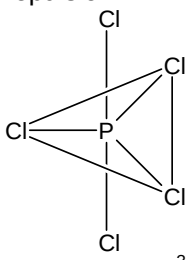
4. $XeOF_4$ is square pyramidal in shape



Due to lone pair-bond pair repulsion the four F atoms move away from lone pair resulting in the movement of Xe towards lone pair.



The most electronegative substituent lies at the axial position to minimize bond pair-bond pair repulsion.



PCl_5 shows sp^3d hybridisation with the involvement of axial dz^2 orbital. Since the axial Cl is at 90° hence suffers relatively greater repulsion increasing the axial bond length.

The size of F is smaller than Cl leading smaller C–F bond length decreasing the dipole moment.

5. Surface tension of lyophilic aquasols is lower than that due to force of attraction between dispered phase and dipersion medium.
Its viscocity is higher due to force of attraction between phase and disperesion medium.

8. White salts or ions giving colourless beads do not give borax bead test.

9. $Cu^{2+} + 2e \longrightarrow Cu$
Hence SO_4^{-2} ions are left which form H_2SO_4 with H^+ from H_2O.

11. At anode: $2OH^- \longrightarrow H_2O + \frac{1}{2}O_2 + 2e^-$

Equivalent volume $V_e = \frac{22.4}{2 \times 2} = 5.6$ litre

$\therefore\ V = \dfrac{ItV_e}{96500} = \dfrac{1\times 965\times 5.6\times 10^3}{96500}$ = **56 ml**.

SECTION –C

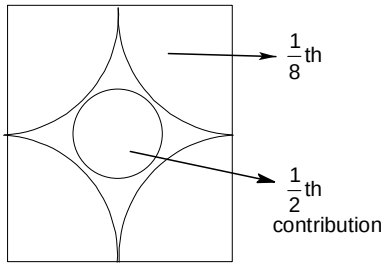
1. At equilibrium relative lowering of vapour pressure of both the solution is same. Then

$$\frac{(1+\alpha)0.01}{60} = \frac{0.014}{60}$$

$(1 + \alpha) = 1.4$
$\alpha = 0.4$
so
$10\alpha = 4$

2. Consider the unit cell face
So, atoms on face will be

$= \dfrac{1}{8}\times\dfrac{1}{4}+\dfrac{1}{2}\times 1 = 1$ atom



No. of atoms at one face of crystal

$= \dfrac{1}{6}\times 6\times 10^{30} = 10^{30}$ atm

No. of unit cells at one face of crystal = $6\times 10^{30}/6$

So no. of unit cells at edge of crystal = $\sqrt{10^{30}} = 10^{15}$

Now edge length of one unit cell = $\dfrac{4}{\sqrt{2}}\times 50\times 10^{15}$ n

So area of face of crystal = $\left(\dfrac{4}{\sqrt{2}}\times 50\times 10^{15}\right)$nm^2

$= \dfrac{16}{2}\times 25\times 10^2\times 10^{30}$
$= 2\times 10^{34}$ nm^2
$= 2\times 10^{-18+34}$ m^2
$= 2\times 10^{16}$ m^2

3. Given P = 1.56 atm, V = 10 L
T = 317 K, R = 0.083 L

Total moles (n) = $\dfrac{PV}{RT} = \dfrac{1.56\times 10}{0.082\times 31\%} = 0.6$ mol

Let C_xH_8 be a mol, therefore moles of C_xH_{12} = (0.6 – a) mol. Mass of C in a mol of C_xH_{12} = 129 x gm.
Mass of C in (0.6 – a) mol of C_xH_{12} = 12 (0.6 – a) gram
Total mass of C in mixture = 12ax + 12x (0.6 – a)g
= 41.4 g

% of C in mixture = $\frac{7.2x}{41.4}\times 100$

Given % of C = 87%

Or

$\frac{720x}{41.4} = 87$

x = 5

4. For $A + B \rightleftharpoons AB$

$K = \frac{[AB]}{[A][B]}$

When $2A + 2B \rightleftharpoons 2AB$

Now eq. conc. will be 2 times of AB

5. $SrF_2 \rightleftharpoons Sr^{+2}(aq) + 2F^{-}(aq)$

$Ksp = \left[Sr^{+2}\right]\left[F^{-}\right]^2$

$\left[Sr^{+2}\right] = \frac{Ksp}{\left[F^{-}\right]^2} = \frac{7.9\times 10^{10}}{\left[4.0\times 10^{-2}\right]^2} = 4.938\times 10^{-7} M$

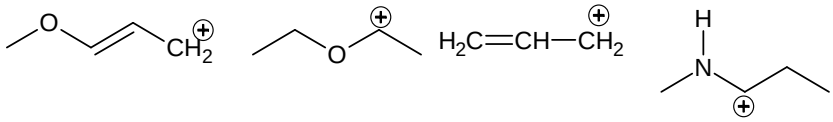
This concentration of Sr^{+2} ions should also satisfy the other expression for $SrCO_3$

$SrCO_3(s) \rightleftharpoons Sr^{+2}(aq) + CO_3^{--}(aq)$

$Ksp = \left[Sr^{+2}\right]\left[CO_3^{--}\right]$

$\left[CO_3^{--}\right] = \frac{Ksp}{\left[Sr^{+2}\right]} = \frac{7.0\times 10^{-10}}{4.938\times 10^{-7}}$

$= 1.4 \times 10^{-3}$

6. 

Mathematics

PART – III

SECTION – A

1. Write the functional equation as $f(x) - f\left(\frac{x}{2}\right) = \frac{x}{2}$

$\Rightarrow f\left(\frac{x}{2}\right) - f\left(\frac{x}{4}\right) = \frac{x}{4}$

$\Rightarrow f\left(\frac{x}{4}\right) - f\left(\frac{x}{8}\right) = \frac{x}{8}$

– – – – – – – – – – –

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$f\left(\frac{x}{2^{n-1}}\right)-f\left(\frac{x}{2^n}\right)=\frac{x}{2^n}$

Adding up we obtain $f(x)-f\left(\frac{x}{2^n}\right)=x\left(\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+.....+\frac{1}{2^n}\right)$

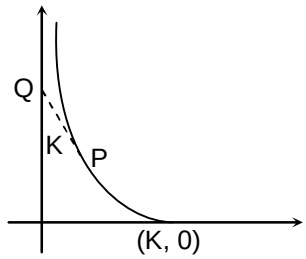
Hence, $\lim_{n\to\infty}\left(f(x)-f\left(\frac{x}{2^n}\right)\right)=x$

2. Differentiating the given equation $2014\,(f(x))^{2013}\,f'(x)=f(x)$
$\Rightarrow 2014\,(f(x))^{2012}\,f'(x)=1$
Integrate $\frac{2014}{2013}(f(x))^{2013}=x+c$
$\because f(0)=1\Rightarrow c=\frac{2014}{2013}$
Hence, $(f(x))^{2013}=\frac{2013}{2014}x+1$
$\Rightarrow (f(2014))^{2013}=2013+1=2014$

3. Let $A_i \equiv$ the event that student correctly solves exactly i of the three proposed problems (Where i = 0, 1, 2, 3)
Required probability = $P(A)=P(A_2\cup A_3)=P(A_2)+P(A_3)$
$\because$ Student know how to solve exactly half the problem
$P(A_0)=P(A_3)$ and $P(A_1)=P(A_2)$
Also, $P(A_0)+P(A_1)+P(A_2)+P(A_3)=1$
$\Rightarrow P(A)=P(A_2)+P(A_3)=\frac{1}{2}$

4. Let r to in-radius and r_1, r_2, r_3 are ex-radii of circle touching the sides BC, CA and AB respectively of ΔABC
If $\Delta$is area of triangle and s is semi-perimeter, $\Delta = rs = r_1(s-a) = r_2(s-b) = r_3(s-c)$
$\Rightarrow \frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3}=\frac{s-a}{\Delta}+\frac{s-b}{\Delta}+\frac{s-c}{\Delta}=\frac{s}{\Delta}=\frac{1}{r}$
So, that r is the smallest term in G.P. Suppose that $A\le B\le C$
then $r_1=s\tan\frac{A}{2}\le r_2=s\tan\frac{B}{2}\le r_3=s\tan\frac{C}{2}$
Hence, there is a number $t>1$ for which $r_1=tr$, $r_2=rt^2$, $r_3=rt^3$
and $(s-b)=t(s-c)$, $(s-a)=t^2(s-c)$, $s=t^3(s-c)$
By Herons formula $t^3r(s-c)=r_3(s-c)=rs=\Delta=\sqrt{s(s-a)(s-b)(s-c)}=\sqrt{t^6(s-c)^4}=t^3(s-c)^2$
Hence, $r=s-c$, and so $\tan\frac{C}{2}=\frac{r}{(s-c)}=1\Rightarrow C=90^o$

5. If the curve traced by P is $y=f(x)$ then $\frac{dy}{dx}$ = slope of line joining P and Q as P is always moving towards Q
Let $Q\equiv(0,\beta)$ and $P(x,y)$, $(0-x)^2+(\beta-y)^2=k^2$
$\Rightarrow \beta-y=\sqrt{K^2-x^2}$
$\frac{dy}{dx}=\frac{\beta-y}{0-x}=\frac{\sqrt{K^2-x^2}}{-x}$
Solving the differential equation with initial condition f(K) = 0 gives



$$y = -K\ln\left(\frac{K+\sqrt{K^2-x^2}}{x}\right) - \sqrt{K^2-x^2}$$

6. Differentiating the given equation $2f(x)\,f'(x) = (f(x))^2 + (f'(x))^2 \Rightarrow (f(x) - f'(x))^2 = 0$
$\Rightarrow f'(x) = f(x) \Rightarrow f(x) = ce^x$
$\because$ f(0) = $\sqrt{K} \Rightarrow c = \sqrt{K} \Rightarrow f(x) = \sqrt{K}e^x$

7. Point of intersection of P_1 and L is given by $mx^2 + \ell x + n = 0$
Line is tangent if $\ell^2 = 4mn \Rightarrow$ m, $\frac{\ell}{2}$, n are in G.P.
If point of intersection is rational (let $x = \frac{p}{q}$) where p and q are co-prime
Then $mp^2 + lpq + nq^2 = 0$ ….. (1)
Now, if one of p and q is even and other is odd then (1) can not hold as sum of an even and an odd integer can't be zero
If p, q are odd then (1) can not hold true as sum of three odd numbers can't be zero
Common tangent to P_1 and P_2 is $2x - y - 1 = 0$
Common chord of P_1 and P_2 is $2x + y = 0$

8. The sequence of two horses reaching the finishing line can be 2 or 1 + 1
$\Rightarrow$ Number of ways = 1 + 2! = 3
The sequence of three horses reaching the finishing line and number of ways are shown below
$3 \Rightarrow$ 1 way
$2 + 1 \Rightarrow {}^3C_2 = 3$ ways
$1 + 2 \Rightarrow {}^3C_2 = 3$ ways
$1 + 1 + 1 \Rightarrow 3! = 6$ ways
Hence $H_3 = 1 + 3 + 3 + 6 = 13$
The sequence of 4 horses reaching the finishing line and number of ways are shown below
$4 \Rightarrow$ 1 way
$2 + 1 + 1 \Rightarrow \left(\frac{4!}{2!\,1!\,1!\,2!}\right) \times 3!$ ways
$3 + 1 \Rightarrow {}^4C_3$ ways
$1 + 3 \Rightarrow {}^4C_1$
$2 + 2 \Rightarrow {}^4C_2$
$1 + 1 + 1 + 1 \Rightarrow 4!$
Hence, $H_4 = 1 + 4 + 4 + 6 + 36 + 24 = 75$

9.-11. Equations of lines L_1 and L_2 can be obtained in symmetric form
L_1: $\frac{x}{0} = \frac{y-b}{-b} = \frac{z}{c}$
L_2: $\frac{x-a}{a} = \frac{y}{0} = \frac{z}{c}$
Equation of plane p: $\begin{vmatrix} x & y-b & z \\ 0 & -b & c \\ a & 0 & c \end{vmatrix} = 0$
$\Rightarrow \frac{x}{a} - \frac{y}{b} - \frac{z}{c} + 1 = 0$
Lines L_1 and L_2 in vector form
$L_1 \equiv \vec{r} = \vec{p}_1 + \lambda\vec{q}_1 \Rightarrow \vec{r} = b\hat{j} + \lambda\left(-b\hat{j} + c\hat{k}\right)$ and $L_2 \equiv \vec{r} = \vec{p}_2 + \lambda\vec{q}_2 \Rightarrow \vec{r} = a\hat{i} + 4\left(a\hat{i} + c\hat{k}\right)$

Shortest distance = $\dfrac{\left[\vec{p}_1-\vec{p}_2\ \ \vec{q}_1\ \ \vec{q}_2\right]}{\left|\vec{q}_1\times\vec{q}_2\right|}=\dfrac{\begin{vmatrix}-a & b & 0\\ 0 & -b & c\\ a & 0 & c\end{vmatrix}}{\sqrt{(bc)^2+(ac)^2+(ab)^2}}=\dfrac{1}{4}$

$\Rightarrow \left|\dfrac{2abc}{\sqrt{(bc)^2+(ca)^2+(ab)^2}}\right|=\dfrac{1}{4} \Rightarrow \dfrac{1}{a^2}+\dfrac{1}{b^2}+\dfrac{1}{c^2}=64$

SECTION – B

1. (A) $f(x) = x[x]$
f(x) is continuous at x = 0, but not differentiable at x = 0
f(x) is discontinuous at integers
(B) $f(x) = x|x|$

$$f(x)=\begin{cases}x^2 & , \ x\ge 0\\ -x^2 & , \ x<0\end{cases}$$

f(x) is continuous and differentiable everywhere
(C) $f(x) = x + |x| = \begin{cases}2x & , \ x\ge 0\\ 0 & , \ x<0\end{cases}$
f(x) is continuous everywhere but not differentiable at x = 0
(D) $f(x) = x = [x]$
f(x) is discontinuous at integers

2. (A) Given lines are
$3x – 4y + 5 = 0$ ….. (1)
$4x + 3y – 10 = 0$ ….. (2)
$x + y + 5 = 0$ ….. (3)
Radical centre of three circles described on sides of Δ as diameter = orthocentre
As two lines are perpendicular to each other
So, orthocentre is point of intersection of (1) and (2)
Solving (1) and (2) x = 1, y = 2
So, radical centre (1, 2)
$4x – ky + 1 = 0$ is the given line (0, 0) 0 + 0 + 1 > 0
So, for radical centre to be on opposite side 4 – 2k + 1 < 0
5 – 2k < 0
$\Rightarrow k > \dfrac{5}{2}$
(B) $x^2 + 2\lambda xy + 2y^2 = 0$
Equation of angle bisectors $\dfrac{x^2-y^2}{-1}=\dfrac{xy}{\lambda}$ ….. (1)
$(1 + \lambda)x^2 – 8xy + y^2 = 0$
Equation of angle bisectors $\dfrac{x^2-y^2}{\lambda}=\dfrac{xy}{-4}$ ….. (2)
For equally inclined lines (1) and (2) must represent same
So, $-\dfrac{1}{\lambda}=\dfrac{\lambda}{-4} \Rightarrow \lambda^2 = 4, \lambda = \pm 2$
(C) $y^2 – 5x + 6y + 15 = 0$

$y^2 + 6y + 9 = 5\left(x - \frac{6}{5}\right)$

$(y+3)^2 = 5\left(x - \frac{6}{5}\right)$

$Y^2 = 4aX,\ a = \frac{5}{4}$

$X = x - \frac{6}{5}$

For three normals to be distinct

$X - 2a > 0$

$x - \frac{6}{5} > \frac{5}{2}$

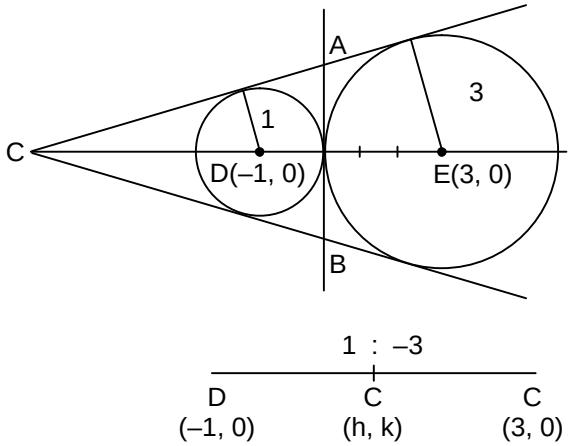
$x > \frac{5}{2} + \frac{6}{5}$

$x > \frac{37}{10}$

(D) $x^2 + y^2 - 6x = 0$

$x^2 + y^2 + 2x = 0$

The triangle formed by common tangents is ΔABC



Let $C \equiv (h, k)$, $\frac{CD}{CE} = \frac{1}{3}$

$h = \frac{3+3}{-2} = -3$

$k = 0$

So, $C \equiv (-3, 0)$

Equation of line thru Q, $y = m(x + 3)$

$\Rightarrow mx - y + 3m = 0$

This represents two tangents with two difference values of m

Length of perpendicular from D = Radius $\frac{|-m+3m|}{\sqrt{m^2+1}} = 1$

$4m^2 = m^2 + 1$

$3m^2 = 1$

$m = \pm\frac{1}{\sqrt{3}}$

So, equation of tangents $y = \frac{x}{\sqrt{3}} + \sqrt{3}$ and $y = \frac{-x}{\sqrt{3}} - \sqrt{3}$

$\Rightarrow A \equiv \left(0, \sqrt{3}\right),\ B \equiv \left(0, -\sqrt{3}\right),\ C \equiv (-3, 0)$

$\Rightarrow AB = BC = CA = 2\sqrt{3}$

So, $P = 6\sqrt{3}$, $A = \frac{\sqrt{3}}{4} \times \left(2\sqrt{3}\right)^2 = 3\sqrt{3}$

$\frac{P}{\sqrt{3}} = 6$, $\frac{A}{\sqrt{3}} = 3$

SECTION – C

1. $f(x) = x^{\alpha} + 3x^{\beta}$
$f'(x) = \alpha\, x^{\alpha-1} + 3\beta x^{\beta-1}$
$\Rightarrow f'(x) > 0\ \forall\ x \in (0, 1)$
$f''(x) = \alpha(\alpha - 1)x^{\alpha-2} + 3\beta(\beta - 1)x^{\beta-2}\ \forall\ x \in (0, 1)$
$\Rightarrow f''(x) > 0\ \forall\ x \in (0, 1)$
Hence, f(x) is monotonically increasing and concave upwards curve
$B = \dfrac{0+1}{2}$, A = 1

2. $T_r = \dfrac{r^2}{(2r-1)(2r+1)} = \dfrac{1}{4}\left(\dfrac{r}{2r-1} + \dfrac{r}{2r+1}\right)$

$\therefore\ S = \dfrac{1}{4}\sum_{r=1}^{500}\left(\dfrac{r}{2r-1} + \dfrac{r}{2r+1}\right)$

$\Rightarrow\ 4S = \sum_{r=1}^{500}\left(\dfrac{r}{2r-1} + \dfrac{r}{2r+1}\right) = \dfrac{1}{1} + \left(\dfrac{1}{3} + \dfrac{2}{3}\right) + \left(\dfrac{2}{5} + \dfrac{3}{5}\right) + \left(\dfrac{3}{7} + \dfrac{4}{7}\right) + \ldots\ldots + \dfrac{500}{999} + \dfrac{500}{1001}$

$\because\ \dfrac{r}{2r+1} + \dfrac{r+1}{2r+1} = 1$

Combining all such terms, we get $4S = 1 + 499 + \dfrac{500}{1001} = \dfrac{501000}{1001}$

$S = \dfrac{125250}{1001} \Rightarrow [S] = 125$
Number of divisors = 4

3. $I(a) = a^2\int_0^{\pi} x^2(\pi - x)^2\,dx - 2a\int_0^{\pi} x(\pi - x)\sin x\,dx + \int_0^{\pi}\sin^2 x\,dx$

$\Rightarrow I(a) = \dfrac{\pi^5}{30}a^2 - 8a + \dfrac{\pi}{2}$

$I(a) = \dfrac{\pi^5}{30}\left(a - \dfrac{120}{\pi^5}\right)^2 + \dfrac{\pi}{2} - \dfrac{480}{\pi^5}$

Hence, I(a) is minimum for $a = \dfrac{120}{\pi^5} \Rightarrow \alpha = 1, \beta = 5$

4. Dividing P(x) by Q(x) gives $P(x) = Q(x)(x^2 + 1) + x^2 - x + 1$
then $\sum_{i=1}^{4} P(z_i) = \left(\sum_{i=1}^{4} z_i\right)^2 - 2\sum z_i z_j - \sum_{i=1}^{4} z_i + 4$

Use the formula for sum and products of the roots to obtain $\sum_{i=1}^{4} P(z_i) = 1 + 2 - 1 + 4 = 6$

5 $\vec{B} \times \vec{C} = x\vec{A} + y\vec{B} + z\vec{C}$

Taking dot product with $\vec{B} \times \vec{C}$

$(\vec{B} \times \vec{C})(\vec{B} \times \vec{C}) = x\left[\vec{A}\ \vec{B}\ \vec{C}\right] \Rightarrow x = \dfrac{\vec{B}^2 \cdot \vec{C}^2 - (\vec{B} \cdot \vec{C})^2}{\left[\vec{A}\ \vec{B}\ \vec{C}\right]}$

Similarly taking dot with $\vec{C}\times\vec{A}$ gives $y=\dfrac{\left(\vec{B}\times\vec{C}\right)\left(\vec{C}\times\vec{A}\right)}{\left[\vec{A}\,\vec{B}\,\vec{C}\right]}$

and dot with $\vec{A}\times\vec{B}$ gives $z=\dfrac{\left(\vec{B}\times\vec{C}\right)\cdot\left(\vec{A}\times\vec{B}\right)}{\left[\vec{A}\,\vec{B}\,\vec{C}\right]}$

$\because\ \left[\vec{A}\,\vec{B}\,\vec{C}\right]=12$

$\Rightarrow\ \vec{B}^2\cdot\vec{C}^2-\left(\vec{B}\cdot\vec{C}\right)=12$

$\Rightarrow x = 1$

$\Rightarrow\ y=\dfrac{1}{2}$ and $z=-\dfrac{1}{2}$

6 $\because\ |aw + b| = 1$

$\Rightarrow |aw + b|^2 = 1 \Rightarrow (aw + b)\left(a\vec{w}+b\right) \Rightarrow a^2 - ab + b^2 = 1$

$\Rightarrow (a - b)^2 + ab = 1$

When $(a - b)^2 = 0$ and $ab = 1$ then (1, 1), (–1, –1)

When $(a - b)^2 = 1$ and $ab = 0$ then (0, 1), (1, 0), (0, –1), (–1, 0)

Hence, total six ordered pairs